Poornash Anandan Sangeetha

(A.S. Poornash)

B.Tech, Computer Science and Engineering

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Education

- **Indian Institute of Technology, Patna** *Bihar, India*
  Bachelor of Technology in Computer Science and Engineering (2021 - 2025)
  – Current **Institute Rank 1** holder in the ongoing Undergraduate Batch of 2021-2025 at IIT Patna with a GPA of **9.76/10.00**
  – Current **Department Rank 1** holder in the Computer Science and Engineering Department at IIT Patna.
  – Nominated for the **President's Gold Medal** due to exceptional academic performance at IIT Patna.

Publications

- **ToxVidLM: A Multimodal Framework for Toxicity Detection in Code-Mixed Videos (Link)**
  *Authors - **A.S. Poornash***, Krishanu Maity*, Sriparna Saha, Pushpak Bhattacharyya*
  – Findings of the Annual Conference of the Association for Computational Linguistics **(ACL), 2024**
- **Seeing Beyond Words: Multimodal Aspect-Level Complaint Detection in Ecommerce Videos (Openreview Link)**
  *Authors - **A.S. Poornash***, Rishikesh Devanathan*, Apoorva Singh*, Sriparna Saha*
  – Main Proceedings of the ACM International Conference on Multimedia **(ACM 'MM), 2024**
- **APTSumm at BioLaySumm Task 1: Biomedical Breakdown, Improving Readability by Relevancy Based Selection (Link)**
  *Authors - **A.S. Poornash**, Atharva Deshmukh, Archit Sharma, Sriparna Saha*
  – Proceedings of the BioNLP workshop of the Association for Computational Linguistics **ACL 2023**.
  – **2nd place** in the readability metric on the **Leaderboard** of **BioLaySum Shared Task**, BioNLP Workshop, ACL 2023
- **ToxVI: a Multimodal LLM-based Framework for Generating Intervention in Toxic Code-Mixed Videos (CIKM '24) (Preprint)**
  *Authors - **A.S. Poornash***, Krishanu maity*, Sriparna Saha and Kitsuchart Pasupa*
  – Short-paper proceedings of the ACM International Conference on Information and Knowledge Management**(CIKM)** 2024
- **HateThaiSent: Sentiment-Aided Hate Speech Detection in Thai Language (Link)**
  *Authors - **A.S. Poornash***, Krishanu Maity*, Shaubhik Bhattacharya, Salisa Phosit, S. Kongsamlit, Sriparna Saha, Kitsuchart Pasupa*
  – IEEE Transactions on Computational Social Systems Journal **(IEEE TCSS), 2024**

∗ **Indicates Equal Contribution**

Scholastic Achievements

- **Academics:** Current **Batch Topper (Institute Rank 1)** in B.Tech 2021 batch of IIT Patna with a **GPA** of **9.76/10.00**
- **IUSSTF-Viterbi '24 :** Scholarship Awardee to conduct research at the **University of Southern California, Viterbi** by the Department of Science & Technology Govt. of India. **One** of only **15** selected students across India. **(Award Letter)**
- **MITACS GRI '24:** Fellowship awardee, University of Alberta, among 600 awardee students from India. **(Award Letter)**
- **DAAD-WISE '24:** Fellowship awardee, TU Darmstadt, among 200 awardee students from India. **(Award Letter)**
- **Amazon ML Challenge'24:** Secured **6th** position among 75000+ participants in the ML based hackathon **(Certificate)**
- **Inter IIT Tech Meet 11.0 at IIT Kanpur '23:** Bagged **First position** among **all IITs** in a **NLP** based **high prep** event DevRev - Expert Answers in a Flash: Improving Domain-Specific Question Answering.
- **SPARK '22:** Selected for the prestigious summer research program of **IIT Roorkee** in Sophomore year.
- **KVPY-SX Fellow '21:** Awarded by IISc and DST, All India Rank **517** among **0.16 million** students and 2000 awardees.
- **JEE Advanced '21:** Secured an All India Rank of **2355** out of **0.225 million** candidates
- **JEE Mains '21:** Stood amongst the **top 0.21 percentile** out of **1 million** candidates

Key Courses Taken

- **Computer Science:** Optimization Techniques (AA) (covers basics of Machine learning and Deep Learning as well), Artificial Intelligence (AA), Machine Learning and Data Science (AB), Data Structures and Algorithms (AA), Database Systems (AA), Computer Architecture (AB), Operating systems (AB), Digital Systems (AA), Compiler Design (AB), Computer Networks (AA), Computer Graphics (AA)
- **Math:** Discrete Mathematics (AA), Linear Algebra (AA), Multivariate Calculus (AA), Probability and Statistics (AA), Real Analysis (AA), Complex Analysis(AA)

Grade Key: AA – 10/10, **AB** – 9/10

Technical Skills

- **Programming/ Tools & OS:** C, C++, Python, MATLAB, Java, Kotlin, Verilog, Arduino, Git, Latex, VS Code, Jupyter Notebooks, Linux, Windows, MySQL, Firebase, Android Studio, Firestore.
- **Frameworks/ Libraries:** Pytorch, Tensorflow, Keras, Caffe, MONAI, Scikit-learn, OpenCV, Scikit-image, Hugging Face transformers, Haystack, Neo4j, TensorFlow Federated, PySyft, Opacus, TensorBoard, FederatedAI/FATE, IBM Differential Privacy Library, Fairlearn, AIF360, NLTK, ITK, Spacy, Scipy, PIL, Napari, Numpy, Pandas
- **Web Skills:** HTML, CSS, Javascript, PHP

Research Experience

- **SteFFLe: A Stochastic Framework for Private, Fair, and Federated Learning (Ongoing Research)** *May. 2024 - July. 2024*
  Guide: Dr. Meisam Razaviyayn | Lab: University of Southern California, ODDS Research Group **Certificate Link**
  - Helped develop a Stochastic **Differentially Private, Fair,** and **Federated** Framework with proven **inter-silo record-level differential privacy** (normally silo-level) with guaranteed convergence for minibatches.
  - Helped prove that the algorithm converges in stochastic settings with **Heterogeneous** data distributions and non-binary attributes.
  - Designed the experiments for the proven results and systematically made **privacy utility tradeoff** curves for various (ϵ, δ) privacy budgets and heterogeneity levels in a federated setting.
  - The proposed method outperforms **SOTA** Private, fair, federated and non-federated frameworks and is **currently under works**.
  - **Skills and Frameworks:** Differential Privacy, Federated Learning, Fairness in AI, Convex and Non-Convex Optimization, Secure Multi-party Computation, Python, PyTorch, TensorFlow, TensorFlow Federated, PySyft, Opacus, TensorBoard, FederatedAI/-FATE, IBM Differential Privacy Library, Fairlearn, AIF360, Scikit-learn, Pandas, NumPy, Jupyter, Docker, Git

- **ToxVidLM: A Multimodal LLM Framework for Toxicity Detection in Code-Mixed Videos** *November. 2023 - May. 2024*
  Guide: Dr. Sriparna Saha | Lab: IIT Patna - AI-NLP-ML Group **Publication Link** **Repository Link**
  - Developed **ToxVidLM**, a **multimodal multitask** framework for **code-mixed video** toxicity detection, achieving **94.29**% accuracy and **94.35**% F1-scores, that has been accepted in **ACL 2024 Findings**.
  - Engineered ToxVidLM with three core modules: Encoder Module for robust multimodal encoding; Cross-Modal Synchronization Module, **aligning** different data modalities with **soft multimodal tokens** generated with **lookup embedding matrices** of language models, and **gated modality fusion**; Multitask Module for managing multiple classification tasks enhancing efficiency and accuracy.
  - **Skills and Frameworks:** Multimodal Deep Learning, Natural Language Processing, Computer Vision, Audio Processing, Python, PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, Librosa, NLTK, Pandas, NumPy, Git.

- **Biomedical Breakdown, Improving Readability by Relevancy Based Selection** *March. 2023 - July. 2023*
  Guide: Dr. Sriparna Saha | Lab: IIT Patna - AI-NLP-ML Group **Publication Link** **Leaderboard Link**
  - Developed a three-step abstractive **summarization** framework for **biomedical documents**, focusing on the lay summary generation that has been accepted to the **BioNLP Workshop** at **ACL 2023**.
  - Participated in the **BioLaySum Shared Task** at the **BioNLP Workshop**, securing the **second-highest readability score** with FKGL at **12.22** and DCRS at **8.99**, demonstrating the effectiveness of our approach in structured lay summarization in scientific content.
  - Employed various NLG metrics for performance evaluation, achieving ROUGE-1 (**48.32**), ROUGE-2 (**14.91**), ROUGE-L (**45.41**).
  - Implemented a **RoBERTa-based re-ranking strategy**, enhancing the quality of candidate summaries generated with beam search sampling through contrastive learning and cosine similarity with the generated RoBERTa embeddings.
  - **Skills and Frameworks:** NLP, Deep learning, Machine Learning, Linguistics, Text preprocessing, Python, PyTorch, Hugging Face Transformers, NLTK, spaCy, NumPy, Pandas, rouge.

- **HateThaiSent: Sentiment-Aided Hate Speech Detection in Thai Language** *July. 2023 - May. 2024*
  Guide: Dr. Kitsuchart Pasupa | King Mongkut's Institute of Technology Ladkrabang, Thailand **Publication Link** **Repository Link**
  - Developed a SOTA **lightweight** multi-task framework combining **FastText** and **mBERT embeddings** with capsule networks, advancing Hatespeech detection in Thai, published in **IEEE Journal of Transactions on Computational Social Sciences**.(*Impact Factor*-**5.0**)
  - Achieved **89.67**% accuracy and **89.79**% macro-F1 score for hate speech; **80.92**% accuracy and **80.97**% macro-F1 score for sentiment analysis, outperforming three SOTA approaches with much fewer parameters in our framework.
  - **Skills and Frameworks:** NLP, Deep Learning, Machine Learning, Python, TensorFlow, Keras, SpaCy, Hugging Face Transformers, TextBlob, Gensim, Git, Data Preprocessing, Scikit-learn, Pandas, NumPy.

- **Median Nerve Detection, Segmentation and Carpal Tunnel Syndrome classfication** *May 2023 - July 2023*
  Guide: Dr. Phaneendra K. Yalavarthy | Lab: IISc Bangalore, Medical Imaging Group (MIG) **Presentation Link** **Certificate Link**
  - Worked on a **medical image segmentation and classification** tasks for Median Nerve detection, segmentation and Carpal Tunnel Syndrome (CTS) diagnosis respectively with an Ultrasound Video Sequences dataset from Aster-CMI hospitals, India.
  - Worked on a dataset of **99 patients**, achieving **Dice scores up to 73.53%** and **IoU improvements up to 68.4%** for MN segmentation.
  - Led development and systematic experiments with **YOLO V3 and UNet models**, notably improving segmentation accuracy in varied configurations, serving as a lightweight alternative to work with in real-time scenarios.
  - **Skills and Frameworks:** Medical imaging, Computer vision, Deep learning, Machine Learning, Image processing, Python, Computational Methods, MATLAB, PyTorch, TensorFlow, OpenCV, MONAI, Scikit-Image, PIL, NumPy, ITK and Napari.

Technical Projects

- **Expert Answers in a Flash: Improving Domain-Specific Question Answering with NLP (Repository link)** *Feb 2023*
  Can answer theme-related questions from a given set of similar theme paragraphs **(Team project at Inter-IIT)**
  - **Skills & Frameworks used**: Natural language processing, Deep Learning, Machine Learning, Python, Pytorch, Tensorflow, Haystack, transformers, Hugging Face, NLTK, numpy, pandas
  - **Techniques used**: Reader-Retriever architecture with theme-specific rankers and heuristics, BM25 retriever, Minilm V2 cross encoder, Minilm Reader, answerability classification with Random Forest classifier
- **The Chandrayaan moon mapping challenge (Repository link)** *Feb 2023*
  Super resolution and custom channel-wise spatial attention models and stitching **(Team project at Inter-IIT)**
  - **Skills & Frameworks used**: Computer vision, Deep Learning, Machine Learning, Python, OpenCV, Pytorch, Numpy
  - **Techniques used**: Cubic interpolation and other image preprocessing techniques, Super-resolution models to increase the spatial resolution, and custom channel-wise spatial attention models

Positions of Responsibility and Extra-Curricular Activities

- **NJACK Machine Learning Divison (Computer Science club of IIT Patna)** *2022- Ongoing*
  Lead IIT Patna
  - **Organised** a Machine learning **hackathon** in collaboration with a startup Chi SquareX during Tech fest of IIT Patna
  - **Conducted talks** by industry professionals, researchers (alumni), **Research Paper reading** and educational sessions on regular basis.
- **National Service Scheme (NSS), IIT Patna** *2021- 2022*
  Volunteer IIT Patna
  - Actively participated in **four** cleanliness drives in various parts of the institute.
  - Taught Maths **weekly 2 hours** in a nearby public primary school as a part of the program
- **Hexachrome (Puzzle Club of IIT Patna)** *2023-2024*
  Sub-Lead IIT Patna
  - **Conducted** various Speedcubing **competitions** with 2x2, 3x3, 4x4, Pyraminx,etc. Designed and released weekly Sudoku puzzles.